

Process Creation Investigation – PowerShell (powershell.exe)

Objective

The objective of this investigation is to verify that PowerShell process execution events are successfully collected by Elastic SIEM using Sysmon logs and to analyze the activity for potential indicators of malicious behavior. PowerShell is a legitimate Windows administration tool but is also frequently abused by attackers for executing scripts, downloading payloads, privilege escalation, and post-exploitation activities.

Command Executed

Get-Process

Detection Method (KQL)

`process.name : "powershell.exe"`

If your environment uses another indexed field, use the query that successfully located the event.

Evidence Collected

Timestamp: 2026-07-15T02:06:59.747Z

Process Name: powershell.exe

Host Name: Sepisecurity

User Domain: Sepisecurity

Process ID (PID): 16692

Event Code: 5 (Verify this matches the actual event source.)

Analysis

The investigation confirmed that the PowerShell process executed successfully on the monitored Windows endpoint. The collected evidence identified the user account, host system, process identifier, and execution timestamp. No indicators of encoded commands, remote script downloads, malicious child processes, or suspicious command-line arguments were observed. Based on the available telemetry, the activity appears consistent with legitimate administrative use of PowerShell.

Security Impact

PowerShell is frequently abused by attackers because it is built into Windows and provides powerful automation capabilities. Monitoring PowerShell execution helps detect malicious scripts, encoded commands, remote execution, malware downloads, and post-exploitation activity. Early detection enables faster incident response and reduces organizational risk.

Conclusion

This investigation demonstrates the ability to monitor Windows process execution using Elastic SIEM and Sysmon, collect forensic evidence, analyze endpoint telemetry, and document findings in a professional SOC investigation format. No malicious activity was identified during this investigation.